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Semiconductor device and method of detecting semiconductor wafer centered on tape

Abstract

A semiconductor manufacturing equipment has a wafer tape including a plurality of alignment holes formed through the wafer tape. A semiconductor wafer is disposed over the wafer tape. The semiconductor wafer includes a circular or rectangular form-factor. A light source is disposed under the wafer tape. The semiconductor wafer is misaligned on the wafer tape with light passing through one or more alignment holes. The semiconductor wafer is centered on the wafer tape with no light passing through one or more alignment holes. The wafer tape has a plurality of wafer alignment markings for different size semiconductor wafers. A light detector is disposed over the semiconductor wafer to detect light passing through the wafer tape. A control arm can be attached to the semiconductor wafer to provide the ability to move the semiconductor wafer in response to a control signal from the light detector.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates in general to semiconductor devices and, more particularly, to a semiconductor device and method of detecting a semiconductor wafer centered on a tape.

BACKGROUND OF THE INVENTION

(2) Semiconductor devices are commonly found in modern electrical products. Semiconductor devices perform a wide range of functions, such as signal processing, high-speed calculations, transmitting and receiving electromagnetic signals, controlling electrical devices, photo-electric, and creating visual images for television displays. Semiconductor devices are found in the fields of communications, power conversion, networks, computers, entertainment, and consumer products. Semiconductor devices are also found in military applications, aviation, automotive, industrial controllers, and office equipment.

(3) A semiconductor wafer typically contains a plurality of semiconductor die separated by a saw street. The semiconductor wafer is commonly mounted to a tape to hold the wafer in place during various manufacturing operations, such as cleaning and dicing. The semiconductor wafer should be centered on the tape for proper alignment with respect to the jig or other dicing tool. The

semiconductor wafer can be moved by hand to attempt to find the center of the tape. However, the manual centering is an estimation at best because the tape is non-transparent. Improper centering can cause imprecise cutting through the wafer and result in lower yield and higher manufacturing cost.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1**a-**1**d illustrate a semiconductor wafer with a plurality of semiconductor die separated by a saw street;

(2) FIGS. **2**a-**21** illustrate a wafer tape with alignment holes to center a semiconductor wafer;

(3) FIG. **3** illustrates the semiconductor wafer subject to a manufacturing operation;

(4) FIG. **4** illustrates a second sized semiconductor wafer centered on the wafer tape by the alignment holes;

(5) FIG. **5** illustrates a third sized semiconductor wafer centered on the wafer tape by the alignment holes;

(6) FIG. **6** illustrates a rectangular semiconductor wafer centered on the wafer tape by the alignment holes;

(7) FIG. **7** illustrates a control arm to move the semiconductor wafer to a center of the wafer tape;

(8) FIGS. **8**a-**8**h illustrate a wafer tape with alignment holes to center a reconstituted wafer; and

(9) FIG. **9** illustrates a printed circuit board (PCB) with different types of packages disposed on a surface of the PCB.

DETAILED DESCRIPTION OF THE DRAWINGS

(10) The present invention is described in one or more embodiments in the following description with reference to the figures, in which like numerals represent the same or similar elements. While the invention is described in terms of the best mode for achieving the invention's objectives, it will be appreciated by those skilled in the art that it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims and their equivalents as supported by the following disclosure and drawings. The features shown in the figures are not necessarily drawn to scale. Elements having a similar function are assigned the same reference number in the figures. The term “semiconductor die” as used herein refers to both the singular and plural form of the words, and accordingly, can refer to both a single semiconductor device and multiple semiconductor devices.

(11) Semiconductor devices are generally manufactured using two complex manufacturing processes: front-end manufacturing and back-end manufacturing. Front-end manufacturing involves the formation of a plurality of die on the surface of a semiconductor wafer. Each die on the wafer contains active and passive electrical components, which are electrically connected to form functional electrical circuits. Active electrical components, such as transistors and diodes, have the ability to control the flow of electrical current. Passive electrical components, such as capacitors, inductors, and resistors, create a relationship between voltage and current necessary to perform electrical circuit functions.

(12) Back-end manufacturing refers to cutting or singulating the finished wafer into the individual semiconductor die and packaging the semiconductor die for structural support, electrical interconnect, and environmental isolation. To singulate the semiconductor die, the wafer is scored and broken along non-functional regions of the wafer called saw streets or scribes. The wafer is singulated using a laser cutting tool or saw blade. After singulation, the individual semiconductor die are disposed on a package substrate that includes pins or contact pads for interconnection with other system components. Contact pads formed over the semiconductor die are then connected to contact pads within the package. The electrical connections can be made with conductive layers,

bumps, stud bumps, conductive paste, or wirebonds. An encapsulant or other molding material is deposited over the package to provide physical support and electrical isolation. The finished package is then inserted into an electrical system and the functionality of the semiconductor device is made available to the other system components.

(13) FIG. **1***a* shows a semiconductor wafer **100** with a base substrate material **102**, such as silicon, germanium, aluminum phosphide, aluminum arsenide, gallium arsenide, gallium nitride, indium phosphide, silicon carbide, or other bulk material for structural support. A plurality of semiconductor die or electrical components **104** is formed on wafer **100** separated by a non-active, inter-die wafer area or saw street **106**. Saw street **106** provides cutting areas to singulate semiconductor wafer **100** into individual semiconductor die **104**. Semiconductor wafer **100** has a circular form-factor. In one embodiment, semiconductor wafer **100** has a width or diameter of 100-450 millimeters (mm).

(14) FIG. **1***b* shows a cross-sectional view of a portion of semiconductor wafer **100**. Each semiconductor die **104** has a back or non-active surface **108** and an active surface **110** containing analog or digital circuits implemented as active devices, passive devices, conductive layers, and dielectric layers formed within the die and electrically interconnected according to the electrical design and function of the die. For example, the circuit may include one or more transistors, diodes, and other circuit elements formed within active surface **110** to implement analog circuits or digital circuits, such as digital signal processor (DSP), application specific integrated circuits (ASIC), memory, or other signal processing circuit. Semiconductor die **104** may also contain IPDs, such as inductors, capacitors, and resistors, for RF signal processing.

(15) An electrically conductive layer **112** is formed over active surface **110** using PVD, CVD, electrolytic plating, electroless plating process, or other suitable metal deposition process. Conductive layer **112** can be one or more layers of aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), silver (Ag), or other suitable electrically conductive material. Conductive layer **112** operates as contact pads electrically connected to the circuits on active surface **110**.

(16) FIG. **2***a* shows backside coating (BSC) or wafer tape **120** made from polymer or epoxy with silver, silica, or alumina filler. FIG. **2***b* is a top view of BSC tape **120**. BSC tape **120** provides support and protection for semiconductor wafer **100**. BSC tape **120** has a thickness of 110 micrometers (mm) and is wider than the maximum wafer application. FIG. **2***c* shows further detail of BSC **120**, including polyethylene terephthalate (PET) release film **121** having a thickness of 38 mm, double-sided tape **123** with PVC base film and having a thickness of 60 mm, thermosetting type backside coating film **125** having a thickness of 25 mm, acrylic release layer **127** having a thickness of 5 mm, and polyolefin base film **129** having a thickness of 80 mm. BSC tape **120** is non-transparent and blocks infrared light (IR).

(17) In FIG. **2***b*, a first wafer alignment circle or marking **122** is shown for a 200 mm diameter wafer. A second wafer alignment circle or marking **124** is shown for a 300 mm diameter wafer. A third wafer alignment circle or marking **126** is shown for a 450 mm diameter wafer. A plurality of alignment openings or holes is formed in BSC tape **120** for each wafer alignment circle by stencil and punch, laser direct ablation (LDA), or etching process. There are generally at least three alignment holes for each wafer alignment circle. Four alignment holes **130***a*, **130***b*, **130***c*, and **130***d* are equally spaced around wafer alignment circuit **122**, i.e., in 90-degree increments. There are four alignment holes **132***a*, **132***b*, **132***c*, and **132***d*, equally spaced around wafer alignment circuit **124**, and six alignment holes **134***a*, **134***b*, **134***c*, **134***d*, **134***e*, and **134***f*, equally spaced around wafer alignment circuit **126**, due to its larger size. The alignment holes **130-134** are typically circular, for a circular wafer, with a diameter of 5.0 mm, although the alignment holes can have other geometric shapes depending on the wafer. Each alignment circle can have any number of alignment holes. The alignment holes **130-134** are formed just inside the wafer alignment circle so that if semiconductor wafer **100** is disposed on the alignment circle, and centered with center **128**, then the alignment circuit would be visible but all alignment holes for that particular alignment circle

would not be visible as they would be covered by the semiconductor wafer. In that case, semiconductor wafer **100** would be considered as ideally aligned with center **128** of BSC tape **120**.

(18) In FIG. **2***d*, semiconductor wafer **100** from FIG. **1***b* is disposed over BSC tape **120**. In the present example, semiconductor wafer **100** is 450 mm in diameter. FIG. **2***e* shows active surface **110** of semiconductor wafer **100** disposed over and in contact with surface **136** of BSC tape **120**. In another embodiment, FIG. **2***f* shows back surface **108** of semiconductor wafer **100** disposed over and in contact with surface **136** of BSC tape **120**. Either orientation of semiconductor wafer **100** is applicable.

(19) FIG. **2***g* shows light source **140** position below BSC tape **120**. Light source **140** emits light waves **142** onto surface **144** of BSC tape **120**. Depending on the position of semiconductor wafer **100**, light waves **142** may or may not transmit through alignment holes **134***a***-134***f*. Assume semiconductor wafer **100** is off-center with respect to center **128**. Again, alignment holes **134***a***-134***f* are formed just inside wafer alignment circle **126**. Any material misalignment of semiconductor wafer **100** with respect to center **128** would allow light waves **142** to pass through one or more alignment holes **134***a***-134***f*. In other words, if misaligned with respect to center **128**, semiconductor wafer **100** will not cover all alignment holes **134***a***-134***f* and light waves **142** will transmit through one or more uncovered alignment holes. In the top view of FIG. **2***h*, semiconductor wafer **100** is off-center with respect to center **128** and semiconductor wafer **100** does not cover alignment holes **134***e* and **134***f*. Light waves **142** will transmit through alignment holes **134***e* and **134***f* and be observable or detectable from the opposite side of BSC tape **120**, i.e., above surface **136**. Light waves **142** can be observable by human sight or detectable by light detector **146**. Light detector **146** can give a visible or audible signal that light waves **142** have been detected above surface **136**, after passing through one or more of alignment holes **134***a***-134***f*. Alignment holes **130***a***-130***d* and **132***a***-132***d* are intended for smaller semiconductor wafers and should not be relevant for a 450 mm semiconductor wafer.

(20) Given that semiconductor wafer **100** is off-center with respect to center **128** in FIG. **2***b*, the position of semiconductor wafer relative to center **128** can then be moved or adjusted until the semiconductor wafer covers all alignment holes **134***a***-134***f*. In that case, no light is observable or detectable through any of the alignment holes **134***a***-134***f*. When no light is observable or detectable through any of the alignment holes **134***a***-134***f*, as in FIG. **2***i*, then semiconductor wafer **100** is considered centered with respect to center **128**.

(21) Once centered, semiconductor wafer **100** can be subject to a variety of semiconductor manufacturing processes. As an example, semiconductor wafer **100** is singulated along saw streets **106**, as shown in FIG. **3**. The alignment of semiconductor wafer **100** with respect to center **128**, as described above, provides precise control over the location of the dicing operation. Alternatively, semiconductor wafer **100** could be subject to pick and place operations, cleaning, and inspection.

(22) FIG. **4** shows semiconductor wafer **100** mounted to BSC tape **120**. In this case, semiconductor wafer **100** is 300 mm in diameter. If semiconductor wafer **100** is off-center with respect to center **128**, light waves **142** pass through one or more alignment holes **132***a***-132***d* inside wafer alignment circle **124**. In other words, if misaligned with respect to center **128**, semiconductor wafer **100** will not cover all alignment holes **132***a***-132***d* and light waves **142** will transmit through the uncovered alignment holes, similar to FIG. **2***h*. Light waves **142** will transmit through the uncovered alignment holes and be observable or detectable from the opposite side of BSC tape **120**, i.e., above surface **136**. Again, light waves **142** can be observable by human sight or detectable by light detector **146**. Light waves **142** are focused on alignment holes **132***a***-132***d* so any light passing through alignment holes **134***a***-134***f* would be negligible or disregarded.

(23) If semiconductor wafer **100** is off-center with respect to center **128**, the position of semiconductor wafer relative to center **128** can then be moved or adjusted until the semiconductor wafer covers all alignment holes **132***a***-132***d*. In that case, no light is observable or detectable through any of the alignment holes **132***a***-132***d*. When no light is observable or detectable through

any of the alignment holes **132***a*-**132***d*, as shown in FIG. **4**, then semiconductor wafer **100** is considered centered with respect to center **128**. Alignment holes **130***a*-**130***d* are intended for a smaller semiconductor wafer and should not be relevant for a 300 mm semiconductor wafer.

(24) FIG. **5** shows semiconductor wafer **100** mounted to BSC tape **120**. In this case, semiconductor wafer **100** is 200 mm in diameter. If semiconductor wafer **100** is off-center with respect to center **128**, light waves **142** pass through one or more alignment holes **130***a*-**130***d* inside wafer alignment circle **122**. In other words, if misaligned with respect to center **128**, semiconductor wafer **100** will not cover all alignment holes **130***a*-**130***d* and light waves **142** will transmit through one or more uncovered alignment holes, similar to FIG. **2***h*. Light waves **142** will transmit through the uncovered alignment holes and be observable or detectable from the opposite side of BSC tape **120**, i.e., above surface **136**. Again, light waves **142** can be observable by human sight or detectable by light detector **146**. Light waves **142** are focused on alignment holes **130***a*-**130***d* so any light passing through alignment holes **134***a*-**134***f* and **132***a*-**132***d* would be negligible or disregarded.

(25) If semiconductor wafer **100** is off-center with respect to center **128**, the position of semiconductor wafer relative to center **128** can then be moved or adjusted until the semiconductor wafer covers all alignment holes **130***a*-**130***d*. In that case, no light is observable or detectable through any of the alignment holes **130***a*-**102***d*. When no light is observable or detectable through any of the alignment holes **130***a*-**130***d*, as shown in FIG. **5**, then semiconductor wafer **100** is considered centered with respect to center **128**.

(26) In another embodiment, semiconductor wafer **100** can be moved or adjusted by control arm **148** in response to light detector **146**, as shown in FIG. **6**. Control arm **148** is attached to semiconductor wafer **100**. If light detector **146** detects light waves **142**, then a control signal is sent to control arm **148** to move semiconductor wafer **100** to make a wafer position adjustment. Given the alignment holes where light is detected, control arm **148** will have information as to where to move semiconductor wafer **100**. For example, if light is detected through alignment holes **134***e*-**134***f*, as in FIG. **2***h*, then light detector **164** sends a control signal to control arm **148** to move semiconductor wafer **100** down with respect to center **128**. Once the adjustment has been made, light detector **146** takes another reading and makes further adjustments in a similar manner, if necessary, until semiconductor wafer **100** covers all alignment holes **134***a*-**134***f* and no light waves **142** are transmitted through BSC **120**.

(27) In another embodiment, semiconductor wafer **200** has a rectangular form-factor. Semiconductor wafer **200** is made similar to FIGS. **1***a*-**1***b*. FIG. **7** shows rectangular semiconductor wafer **200** disposed over BSC tape **120**. In this case, the wafer alignment markings would be rectangular. Semiconductor wafer **200** follows a similar description as FIGS. **2**-**6**, with a rectangular form-factor. If semiconductor wafer **200** is misaligned, the light waves will transmit through one or more alignment holes and corrective action is taken, similar to FIGS. **2***h*-**2***i*. If semiconductor wafer **200** is properly aligned, then no light waves will pass through any of the alignment holes of BSC tape **120** and the semiconductor wafer is considered properly centered.

(28) Returning to FIG. **1***c*, an electrically conductive bump material is deposited over conductive layer **112** using an evaporation, electrolytic plating, electroless plating, ball drop, or screen printing process. The bump material can be Al, Sn, Ni, Au, Ag, Pb, Bi, Cu, solder, and combinations thereof, with an optional flux solution. For example, the bump material can be eutectic Sn/Pb, high-lead solder, or lead-free solder. The bump material is bonded to conductive layer **112** using a suitable attachment or bonding process. In one embodiment, the bump material is reflowed by heating the material above its melting point to form balls or bumps **114**. In one embodiment, bump **114** is formed over an under bump metallization (UBM) having a wetting layer, barrier layer, and adhesive layer. Bump **114** can also be compression bonded or thermocompression bonded to conductive layer **112**. Bump **114** represents one type of interconnect structure that can be formed over conductive layer **112**. The interconnect structure can also use bond wires, conductive paste, stud bump, micro bump, or other electrical interconnect.

(29) In FIG. **1***d*, semiconductor wafer **100** is singulated through saw street **106** using a saw blade or laser cutting tool **118** into individual semiconductor die **104**. The individual semiconductor die **104** can be inspected and electrically tested for identification of known good die or known good unit (KGD/KGU) post singulation.
(30) FIG. **8***a* shows a temporary substrate or carrier **150** containing sacrificial base material, such as silicon, polymer, beryllium oxide, glass, or other suitable low-cost, rigid material for structural support. Substrate **150** has major surface **152** and major surface **154**, opposite surface **152**. In one embodiment, carrier **150** is a support structure with a temporary bonding layer **158** formed over the carrier. Temporary bonding layer **158** can be a film or foil bonded to surface **152**.
(31) In FIG. **8***b*, electrical components **156***a***-156***c* are disposed on surface **152** of substrate **150**. Electrical components **156***a***-156***c* can be similar to, or made similar to, semiconductor die **104** from FIG. **1***d* with bumps **114** oriented toward surface **152** of substrate **150**. Alternatively, electrical components **156***a***-156***c* can include other semiconductor die, semiconductor packages, surface mount devices, RF components, discrete electrical devices, or integrated passive devices (IPD).
(32) Electrical components **156***a***-156***c* are positioned over substrate **150** using a pick and place operation. Electrical components **156***a***-156***b* are brought into contact with bonding layer **158**. FIG. **8***c* illustrates electrical components **156***a***-156***c* bonded to substrate **150**, as a reconstituted wafer level package (WLP) **160**. FIG. **8***d* is a top view of reconstituted WLP **160** with electrical components **156***a***-156***c* bonded to substrate **150**. In the present example, reconstituted WLP **160** is 450 mm in diameter.
(33) In FIG. **8***e*, reconstituted WLP **160** is disposed over BSC tape **120**, similar to FIG. **2***d*. Components having a similar function are assigned the same reference number. FIG. **8***f* shows surface **154** of reconstituted WLP **160** disposed over and in contact with surface **136** of BSC tape **120**.
(34) Light source **162** is positioned below BSC tape **120**. Light source **162** emits light waves **164** onto surface **144** of BSC tape **120**. Depending on the position of reconstituted WLP **160**, light waves **164** may or may not transmit through alignment holes **134***a***-134***f*. Assume reconstituted WLP **160** is off-center with respect to center **128**, as shown in FIG. **8***g*. Again, alignment holes **134***a***-134***f* are formed just inside wafer alignment circle **126**. Any material misalignment of reconstituted WLP **160** with respect to center **128** would allow light waves **164** to pass through one or more alignment holes **134***a***-134***f*. In other words, if misaligned with respect to center **128**, reconstituted WLP **160** will not cover all alignment holes **134***a***-134***f* and light waves **164** will transmit through one or more uncovered alignment holes. In this case, light waves **164** transmitting through alignment holes **134***e* and **134***f* will be observable or detectable from the opposite side of BSC tape **120**, i.e., above surface **136**. Light waves **164** can be observable by human sight or detectable by light detector **166**. Light detector **166** can give a visible or audible signal that light waves **164** have been detected above surface **136**, after passing through one or more of alignment holes **134***a***-134***f*.
(35) Given that reconstituted WLP **160** is off-center with respect to center **128**, the position of the reconstituted WLP **160** relative to center **128** can then be moved or adjusted until the reconstituted WLP **160** covers all alignment holes **134***a***-134***f*. In that case, no light is observable or detectable through any of the alignment holes **134***a***-134***f*. When no light is observable or detectable through any of the alignment holes **134***a***-134***f*, as in FIG. **8***h*, then reconstituted WLP **160** is considered centered with respect to center **128**.
(36) Once centered, semiconductor wafer **160** can be subject to a variety of semiconductor manufacturing processes. As an example, semiconductor wafer **100** is singulated along saw streets **106**, similar to FIG. **3**. The alignment of semiconductor wafer **100** with respect to center **128**, as described above, provides precise control over the location of the dicing operation.
(37) FIG. **9** illustrates electrical device **400** having a chip carrier substrate or PCB **402** with a plurality of semiconductor packages disposed on a surface of PCB **402**, including semiconductor

die **104**. Electrical device **400** can have one type of semiconductor package, or multiple types of semiconductor packages, depending on the application.
(38) Electrical device **400** can be a stand-alone system that uses the semiconductor packages to perform one or more electrical functions. Alternatively, electrical device **400** can be a subcomponent of a larger system. For example, electrical device **400** can be part of a tablet, cellular phone, digital camera, communication system, or other electrical device. Alternatively, electrical device **400** can be a graphics card, network interface card, or other signal processing card that can be inserted into a computer. The semiconductor package can include microprocessors, memories, ASIC, logic circuits, analog circuits, RF circuits, discrete devices, or other semiconductor die or electrical components. Miniaturization and weight reduction are essential for the products to be accepted by the market. The distance between semiconductor devices may be decreased to achieve higher density.
(39) In FIG. **9**, PCB **402** provides a general substrate for structural support and electrical interconnect of the semiconductor packages disposed on the PCB. Conductive signal traces **404** are formed over a surface or within layers of PCB **402** using evaporation, electrolytic plating, electroless plating, screen printing, or other suitable metal deposition process. Signal traces **404** provide for electrical communication between each of the semiconductor packages, mounted components, and other external system components. Traces **404** also provide power and ground connections to each of the semiconductor packages.
(40) In some embodiments, a semiconductor device has two packaging levels. First level packaging is a technique for mechanically and electrically attaching the semiconductor die to an intermediate substrate. Second level packaging involves mechanically and electrically attaching the intermediate substrate to the PCB. In other embodiments, a semiconductor device may have the first level packaging where the die is mechanically and electrically disposed directly on the PCB. For the purpose of illustration, several types of first level packaging, including bond wire package **406** and flipchip **408**, are shown on PCB **402**. Additionally, several types of second level packaging, including ball grid array (BGA) **410**, bump chip carrier (BCC) **412**, land grid array (LGA) **416**, multi-chip module (MCM) or SIP module **418**, quad flat non-leaded package (QFN) **420**, quad flat package **422**, embedded wafer level ball grid array (eWLB) **424**, and wafer level chip scale package (WLCSP) **426** are shown disposed on PCB **402**. In one embodiment, eWLB **424** is a fan-out wafer level package (Fo-WLP) and WLCSP **426** is a fan-in wafer level package (Fi-WLP). Depending upon the system requirements, any combination of semiconductor packages, configured with any combination of first and second level packaging styles, as well as other electrical components, can be connected to PCB **402**. In some embodiments, electrical device **400** includes a single attached semiconductor package, while other embodiments call for multiple interconnected packages. By combining one or more semiconductor packages over a single substrate, manufacturers can incorporate pre-made components into electrical devices and systems. Because the semiconductor packages include sophisticated functionality, electrical devices can be manufactured using less expensive components and a streamlined manufacturing process. The resulting devices are less likely to fail and less expensive to manufacture resulting in a lower cost for consumers.
(41) While one or more embodiments of the present invention have been illustrated in detail, the skilled artisan will appreciate that modifications and adaptations to those embodiments may be made without departing from the scope of the present invention as set forth in the following claims.

Claims

1. A semiconductor manufacturing equipment, comprising: a wafer tape configured to receive a semiconductor wafer thereon, the wafer tape including a plurality of alignment holes formed through the wafer tape; and a light source disposed under the wafer tape, wherein the plurality of

alignment holes and the light source are spatially arranged and configured to indicate a misaligned position of the semiconductor wafer on the wafer tape with light passing through one or more alignment holes.
2. The semiconductor manufacturing equipment of claim 1, wherein the plurality of alignment holes and the light source are spatially arranged and configured to indicate a centered position of the semiconductor wafer on the wafer tape with no light passing through the one or more alignment holes.
3. The semiconductor manufacturing equipment of claim 1, wherein the wafer tape includes a plurality of wafer alignment markings for different size semiconductor wafers.
4. The semiconductor manufacturing equipment of claim 3, wherein the alignment holes are located inside the wafer alignment markings, respectively.
5. The semiconductor manufacturing equipment of claim 1, further including a light detector disposed over the semiconductor wafer to detect light passing through the wafer tape.
6. The semiconductor manufacturing equipment of claim 1, wherein the semiconductor wafer includes a circular or rectangular form-factor.
7. A semiconductor manufacturing equipment, comprising: a wafer tape configured to receive a semiconductor wafer thereon, the wafer tape including a plurality of alignment holes formed through the wafer tape; and a light source disposed under the wafer tape, wherein the plurality of alignment holes and the light source are spatially arranged and configured to indicate a misaligned position of the semiconductor wafer on the wafer tape with no light passing through one or more alignment holes.
8. The semiconductor manufacturing equipment of claim 7, wherein the plurality of alignment holes and the light source are spatially arranged and configured to indicate a centered position of the semiconductor wafer on the wafer tape with light passing through the one or more alignment holes.
9. The semiconductor manufacturing equipment of claim 7, wherein the wafer tape includes a plurality of wafer alignment markings for different size semiconductor wafers.
10. The semiconductor manufacturing equipment of claim 9, wherein the alignment holes are located inside the wafer alignment markings, respectively.
11. The semiconductor manufacturing equipment of claim 7, further including a light detector disposed over the semiconductor wafer to detect light passing through the wafer tape.
12. The semiconductor manufacturing equipment of claim 11, further including a control arm attached to the semiconductor wafer to provide the ability to move the semiconductor wafer in response to a control signal from the light detector.
13. The semiconductor manufacturing equipment of claim 7, wherein the semiconductor wafer includes a circular or rectangular form-factor.
14. A method of making a semiconductor device, comprising: providing a wafer tape including a plurality of alignment holes formed through the wafer tape; disposing a semiconductor wafer over the wafer tape; and disposing a light source under the wafer tape, wherein the semiconductor wafer is misaligned on the wafer tape when light passes through one or more alignment holes.
15. The method of claim 14, wherein the semiconductor wafer is centered on the wafer tape when no light passes through the one or more alignment holes.
16. The method of claim 14, wherein the wafer tape includes a plurality of wafer alignment markings for different size semiconductor wafers.
17. The method of claim 16, wherein the alignment holes are located inside the wafer alignment markings, respectively.
18. The method of claim 14, further including disposing a light detector over the semiconductor wafer to detect light passing through the wafer tape.
19. The method of claim 14, wherein the semiconductor wafer includes a circular or rectangular form-factor.

20. A method of making a semiconductor device, comprising: providing a wafer tape including a plurality of alignment holes formed through the wafer tape; disposing a semiconductor wafer over the wafer tape to cover the alignment holes; and disposing a light source under the wafer tape, wherein the semiconductor wafer is centered on the wafer tape when no light passes through one or more alignment holes.
21. The method of claim 20, wherein the semiconductor wafer is misaligned on the wafer tape when light passes through the one or more alignment holes.
22. The method of claim 20, wherein the wafer tape includes a plurality of wafer alignment markings for different size semiconductor wafers.
23. The method of claim 20, further including disposing a light detector over the semiconductor wafer to detect light passing through the wafer tape.
24. The method of claim 23, further including attaching a control arm to the semiconductor wafer to provide the ability to move the semiconductor wafer in response to a control signal from the light detector.
25. The method of claim 20, wherein the semiconductor wafer includes a circular or rectangular form-factor.